

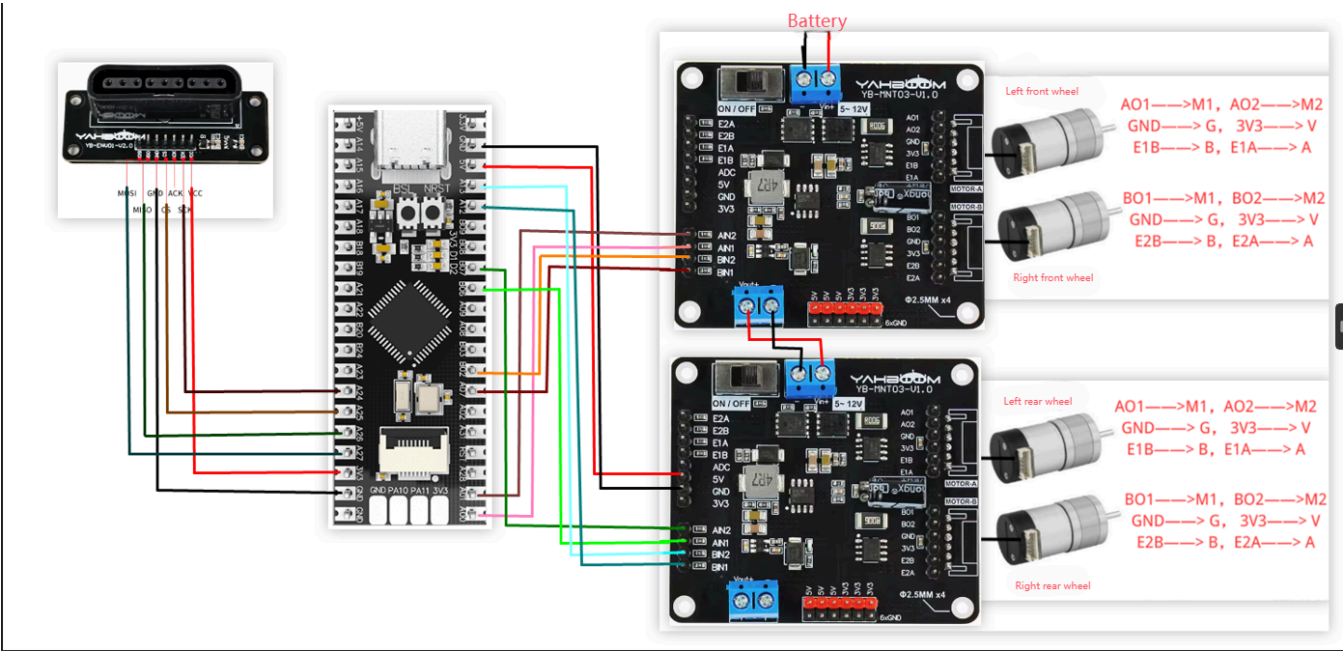
Car control

1. Learning objectives

Control the movement of the car through the handle

2. Hardware connection

Wiring of the handle receiver, MSPM0G3519 and dual-channel motor driver board



Lower driver board (left front wheel, right front wheel)	MSPM0G3519	Upper driver board (left rear wheel, right rear wheel)	MSPM0G3519
AIN1	PA0	AIN1	PB6
AIN2	PA1	AIN2	PB7
BIN1	PA7	BIN1	PA12
BIN2	PB2	BIN2	PA13
PS2 controller receiver	MSPM0G3519	5V	5V
MOSI	PA27	GND	GND
MISO	PA26		
GND	GND		
CS	PA25		
ACK	No connection		

Lower driver board (left front wheel, right front wheel)	MSPM0G3519		Upper driver board (left rear wheel, right rear wheel)	MSPM0G3519
SCK	PA24			
VCC	3V3			

3. Program description

- pstwo.c

```
//判断是否为红灯模式 Judge whether it is red light mode
//返回值: 0, 红灯模式 Return value: 0, red light mode
//其他, 其他模式 Others, other modes
u8 PS2_RedLight(void)
{
    CS_L;
    PS2_Cmd(Comd[0]); //开始命令 Start Command
    PS2_Cmd(Comd[1]); //请求数据 Request data
    CS_H;
    if( Data[1] == 0x73) return 0 ;
    else return 1;
}

void Get_PS2data(void)
{
    PS2_LX=PS2_AnalogData(PSS_LX);
    PS2_LY=PS2_AnalogData(PSS_LY);
    PS2_RX=PS2_AnalogData(PSS_RX);
    PS2_RY=PS2_AnalogData(PSS_RY);
    PS2_KEY=PS2_DataKey();

    // printf("PS2_LX=%d ",PS2_LX);
    // printf("PS2_LY=%d ",PS2_LY);
    // printf("PS2_RX=%d ",PS2_RX);
    // printf("PS2_RY=%d ",PS2_RY);
    // printf("PS2_KEY=%d \r\n",PS2_KEY);
}

void PS2_CarControl(void)
{
    //----摇杆直向控制---//
    //----Joystick vertical control---//
    if(PS2_LX == 128 && PS2_LY == 127)
    {
        Motion_Set_Pwm(0,0,0,0);
    }

    if(PS2_LY>=0&&PS2_LY<70)//直走 Go straight
```

```

{
    Motion_Set_Pwm(200,200,200,200);
}

if(PS2_LY<=255 && PS2_LY>200 &&PS2_LX == 128)//后退    Back
{
    Motion_Set_Pwm(-200,-200,-200,-200);
}

if(PS2_LX>=0&&PS2_LX<70)//左旋    Rotate Left
{
    Motion_Set_Pwm(-400,700,-400,700);
}

if(PS2_LX<=255 && PS2_LX>200 && PS2_LY == 127)//右旋    Rotate Right
{
    Motion_Set_Pwm(700,-400,700,-400);
}

//----摇杆斜向控制---//
//----Joystick tilt control---//
if(PS2_LX >= 0&&PS2_LX <= 90 && PS2_LY >= 0&&PS2_LY <= 70)//左前旋转    Left front
rotation
{
    Motion_Set_Pwm(0,500,0,500);
}
if(PS2_LX >= 165&&PS2_LX <= 255 && PS2_LY >= 0&&PS2_LY <= 70)//右前旋转    Right front
rotation
{
    Motion_Set_Pwm(500,0,500,0);
}
if(PS2_LX >= 0&&PS2_LX <= 90 && PS2_LY >= 165&&PS2_LY <= 255)//左后旋转    Left rear
rotation
{
    Motion_Set_Pwm(0,-500,0,-500);
}
if(PS2_LX >= 165&&PS2_LX <= 255 && PS2_LY >= 165&&PS2_LY <= 255)//右后旋转    Right rear
rotation
{
    Motion_Set_Pwm(-500,0,-500,0);
}

//---方向按键控制---//
//---Direction button control---//
switch(PS2_KEY){
    case KEY_FORWARD:
        Motion_Set_Pwm(200,200,200,200);
        break;

    case KEY_BACK:
        Motion_Set_Pwm(-200,-200,-200,-200);
        break;
}

```

```

        case KEY_SPINRIGHT:
            Motion_Set_Pwm(700, -400, 700, -400);
            break;

        case KEY_SPINLEFT:
            Motion_Set_Pwm(-400, 700, -400, 700);
            break;

        default:
            break;
    }
}

```

- PS2_RedLight: Realize the red light mode judgment. When the handle is in non-red light mode, that is, traffic light mode, the car can be controlled to move;
- Get_PS2data: Get the joystick analog value and button value of the controller
- PS2_CarControl: Control the car through the acquired data

- bsp_motor.h

```

#define PWM_M1_A(value)
DL_TimerG_setCaptureCompareValue(PWM_0_INST, value, GPIO_PWM_0_C0_IDX);
#define PWM_M1_B(value)
DL_TimerG_setCaptureCompareValue(PWM_0_INST, value, GPIO_PWM_0_C1_IDX);

#define PWM_M2_A(value)
DL_TimerG_setCaptureCompareValue(PWM_0_INST, value, GPIO_PWM_0_C2_IDX);
#define PWM_M2_B(value)
DL_TimerG_setCaptureCompareValue(PWM_0_INST, value, GPIO_PWM_0_C3_IDX);

#define PWM_M3_A(value)
DL_TimerG_setCaptureCompareValue(PWM_1_INST, value, GPIO_PWM_1_C0_IDX);
#define PWM_M3_B(value)
DL_TimerG_setCaptureCompareValue(PWM_1_INST, value, GPIO_PWM_1_C1_IDX);

#define PWM_M4_A(value)
DL_TimerG_setCaptureCompareValue(PWM_2_INST, value, GPIO_PWM_2_C0_IDX);
#define PWM_M4_B(value)
DL_TimerG_setCaptureCompareValue(PWM_2_INST, value, GPIO_PWM_2_C1_IDX);

```

Define the set duty cycle functions for the 4 motor pins.

- bsp_motor.c

```

// 设置电机速度, speed:±1000, 0为停止 // Set the motor speed, speed: ±1000, 0 means stop
void Motor_Set_Pwm(uint8_t id, int16_t speed)
{
    int16_t pulse = Motor_Ignore_Dead_Zone(speed);
    // 限制输入 Limit Input
    if (pulse >= MOTOR_MAX_PULSE)

```

```

    pulse = MOTOR_MAX_PULSE;
if (pulse <= -MOTOR_MAX_PULSE)
    pulse = -MOTOR_MAX_PULSE;

switch (id)
{
case MOTOR_ID_M1:
{
    if (pulse >= 0)
    {
        PWM_M1_A(pulse);
        PWM_M1_B(0);
    }
    else
    {
        PWM_M1_A(0);
        PWM_M1_B(-pulse);
    }
    break;
}
case MOTOR_ID_M2:
{
    pulse = -pulse;
    if (pulse >= 0)
    {
        PWM_M2_A(pulse);
        PWM_M2_B(0);
    }
    else
    {
        PWM_M2_A(0);
        PWM_M2_B(-pulse);
    }
    break;
}

case MOTOR_ID_M3:
{
    if (pulse >= 0)
    {
        PWM_M3_A(pulse);
        PWM_M3_B(0);
    }
    else
    {
        PWM_M3_A(0);
        PWM_M3_B(-pulse);
    }
    break;
}
case MOTOR_ID_M4:
{
    pulse = -pulse;

```

```

        if (pulse >= 0)
        {
            PWM_M4_A(pulse);
            PWM_M4_B(0);
        }
        else
        {
            PWM_M4_A(0);
            PWM_M4_B(-pulse);
        }
        break;
    }

    default:
        break;
}
}

```

Give the input value to the motor to control the forward and reverse rotation and speed of the motor. This control logic is not suitable for all motors and motor driver boards. If you are using the motor driver board and motor mentioned in this tutorial, if the direction of the car is not correct, you need to check the wiring again. If you are using a motor driver board and motor not mentioned in this tutorial, you need to modify it according to your own situation.

- app_motor.c

```

// 控制小车运动, Motor_x=[-1000, 1000], 超过范围则无效。
// Control the movement of the car, Motor_x=[-1000, 1000]. It will be invalid if it exceeds
the range.
void Motion_Set_Pwm(int16_t Motor_1, int16_t Motor_2, int16_t Motor_3, int16_t Motor_4)
{
    if (Motor_1 >= -MOTOR_MAX_PULSE && Motor_1 <= MOTOR_MAX_PULSE)
    {
        Motor_Set_Pwm(MOTOR_ID_M1, Motor_1);
    }
    if (Motor_2 >= -MOTOR_MAX_PULSE && Motor_2 <= MOTOR_MAX_PULSE)
    {
        Motor_Set_Pwm(MOTOR_ID_M2, Motor_2);
    }
    if (Motor_3 >= -MOTOR_MAX_PULSE && Motor_3 <= MOTOR_MAX_PULSE)
    {
        Motor_Set_Pwm(MOTOR_ID_M3, Motor_3);
    }
    if (Motor_4 >= -MOTOR_MAX_PULSE && Motor_4 <= MOTOR_MAX_PULSE)
    {
        Motor_Set_Pwm(MOTOR_ID_M4, Motor_4);
    }
}

```

Limit the motor input value to not exceed the limit range.

- empty.c

```
int main(void)
{
    extern int AnalogFLAG;

    USART_Init();
    while(1)
    {
        Get_PS2data();
        if(!PS2_RedLight())//切换为模拟量输入模式时，开始控制小车 when switching to analog input
mode, start controlling the car
        {
            PS2_CarControl();
        }
        delay_ms(2);
    }
}
```

After powering on and switching to traffic light mode, start controlling the car's movement.

Note: Be careful when changing the delay function used in this project! Random modification may cause excessive delay or disconnection of the PS2 controller.

Note: The project source code must be placed in the SDK path for compilation.

** For example, the path: D:\TI\M0_SDK\mspm0_sdk_1_30_00_03\1.TB6612**

> 新加卷 (D:) > TI > M0_SDK > mspm0_sdk_1_30_00_03				
名称	修改日期	类型	大小	
1.TB6612	2024/7/22 18:59	文件夹		
2.AT8236	2024/7/22 19:47	文件夹		
3.Enconder	2024/7/23 10:36	文件夹		
4.Servo	2024/7/23 11:13	文件夹		
docs	2024/7/23 10:33	文件夹		
examples	2024/7/23 10:34	文件夹		
kernel	2024/7/23 10:37	文件夹		
source	2024/7/23 10:33	文件夹		
tools	2024/7/23 10:33	文件夹		
imports.mak	2024/1/25 11:45	MAK 文件	2 KB	
known_issues_FAQ.html	2024/1/25 11:42	Microsoft Edge ...	67 KB	
license_mspm0_sdk_1_30_00_03.txt	2024/1/25 11:42	文本文档	33 KB	
manifest_mspm0_sdk_1_30_00_03.html	2024/1/25 11:42	Microsoft Edge ...	113 KB	
mspm0sdk_1_30_00_03.log	2024/7/23 10:42	文本文档	5,237 KB	
release_notes_mspm0_sdk_1_30_00_0...	2024/1/25 11:42	Microsoft Edge ...	108 KB	
uninstall.dat	2024/7/23 10:39	DAT 文件	344 KB	
uninstall.exe	2024/7/23 10:39	应用程序	6,048 KB	

4. Experimental phenomenon

Burn the line patrol program into MSPM0G3519. Patiently connect the wires according to the wiring diagram. After connecting the wires, patiently check whether the wires are connected correctly. If you do not check, the car may not move at best, or the board may be burned directly. After confirming that everything is correct, turn on the upper and lower drive board switches. Then turn on the handle switch, wait until the indicator light of the handle stops flashing, and the red and green lights of the handle receiver are always on. Press the mode button of the handle to switch to the traffic light mode of the handle. Then you can press the directional key and joystick on the left side of the handle to control the car. The directional key up and down controls forward and backward, and the left and right controls left and right rotation. The joystick can control the car forward and backward, diagonally controls left and right rotation, and left and right controls left and right rotation.